

ECON 5345 Homework 7 Report

Harlly Zhou

March 27, 2026

Question 1

- a. The responses and CIs are reported in Figure 1. For the ease of comparison, part (b) is plotted on the same figure. Codes in “hw7_q1.R”.

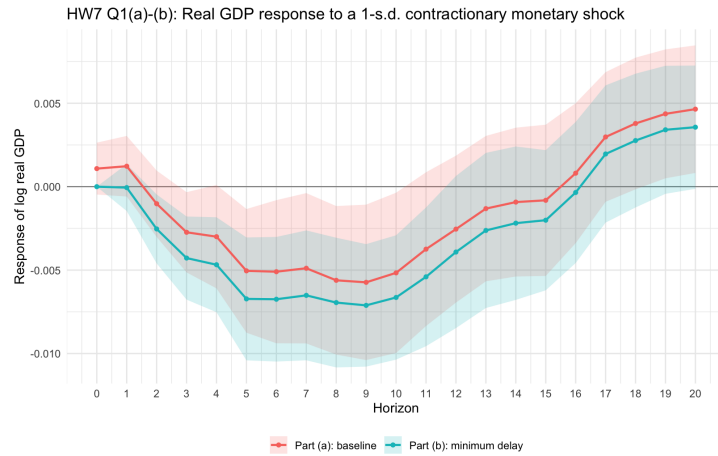


Figure 1: GDP IRF and CI

- b. See Figure 1.

c. The responses and CIs are reported in Figure 2. Codes in “hw7_q1.R”.

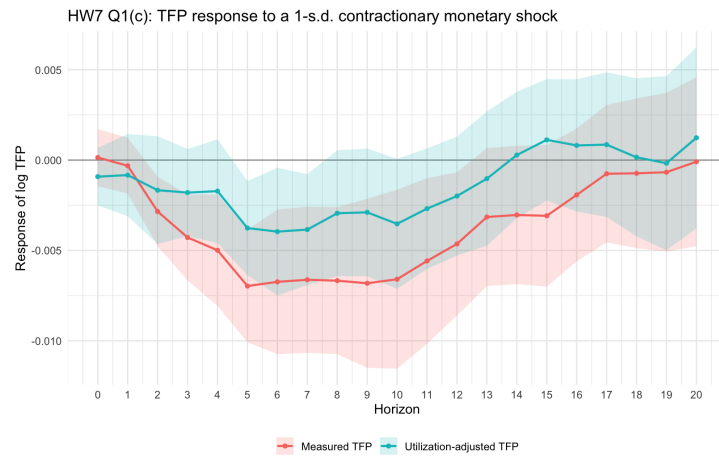


Figure 2: TFP IRF and CI

The measured TFP response has a delayed but statistically significant response to the monetary policy shock after around 2 quarters. The utilization-adjusted TFP has almost no statistically significant response to the monetary policy shock. Since TFP should not endogenously respond to monetary policy shock, this suggests that the utilization-adjusted TFP is a better measure of TFP.

Question 2

a. We have $z_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \varepsilon_t \\ \varepsilon_{t-1} \end{pmatrix}$. Then we can write

$$z_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \varepsilon_t \\ \varepsilon_{t-1} \end{pmatrix} = \begin{pmatrix} \alpha & 0 & \theta & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} y_{t-1} \\ y_{t-2} \\ \varepsilon_{t-1} \\ \varepsilon_{t-2} \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \varepsilon_t$$

and $y_t = (1 \ 0 \ 0 \ 0)z_t$.

b. The following table reports the estimates:

T	$\hat{\alpha}$	$\hat{\theta}$	$\hat{\sigma}^2$	$\text{se}(\hat{\alpha})$	$\text{se}(\hat{\theta})$	$\text{se}(\hat{\sigma})$	LogLik
50	0.5895	0.4642	2.1740	0.1316	0.1523	0.1510	-101.7544
100	0.7960	0.4234	2.5275	0.0693	0.1106	0.1137	-199.3829
500	0.7835	0.3898	3.0852	0.0306	0.0466	0.0557	-1001.9500

Table 1: Estimation results for different sample sizes

c. I first present the statistics for σ^2 because in HR and MoM I did not report them.

T	Bias	Variance	MSE	SD
50	-0.0314	0.3892	0.3898	0.624
100	-0.0469	0.1802	0.1822	0.425
500	-0.0186	0.0331	0.0334	0.182

Table 2: Estimation performance for σ^2 (Kalman Filter).

The comparison of the estimates are reported in Table 3. For α estimation, the Kalman filter has better performance especially when the sample size is small. For θ estimation, the Kalman filter has better performance when the sample size is large.

T	Parameter	Bias	Variance	MSE	SD
Kalman Filter					
50	α	0.0072	0.0109	0.0109	0.104
50	θ	-0.1175	0.0654	0.0792	0.256
100	α	-0.0011	0.0052	0.0052	0.072
100	θ	-0.0479	0.0240	0.0263	0.155
500	α	-0.0026	0.0009	0.0009	0.030
500	θ	-0.0011	0.0021	0.0021	0.046
HR					
50	α	-0.09	—	0.02	0.13
50	θ	0.01	—	0.02	0.15
100	α	-0.04	—	0.01	0.08
100	θ	-0.00	—	0.01	0.11
500	α	-0.01	—	0.00	0.03
500	θ	-0.00	—	0.00	0.05
MoM					
50	α	-0.05	—	0.02	0.12
50	θ	0.06	—	0.06	0.24
100	α	-0.02	—	0.01	0.08
100	θ	0.04	—	0.04	0.19
500	α	-0.01	—	0.00	0.03
500	θ	0.01	—	0.00	0.07

Table 3: Comparison of estimation performance for α and θ .